

QEB → Módulo 4 : Nomenclatura e funções orgânicas
↳ Bruce: 3.1 a 3.6

⇒ Nomenclatura de alcamos (C_nH_{2n+2}) : base para as outras nomenclaturas ↳ hidrocarbonetos

Para alcamos de cadeia reta : prefixo + "ano"

Carbonos	Nome	Fórmula	Estrutura
1	Metano	CH_4	CH_4
2	Etano	C_2H_6	H_3C-CH_3
3	Propano	C_3H_8	
4	Butano	C_4H_{10}	
5	Pentano	C_5H_{12}	
6	Hexano	C_6H_{14}	
7	Heptano	C_7H_{16}	
8	Octano	C_8H_{18}	
9	Nonano	C_9H_{20}	
10	Decano	$C_{10}H_{22}$	

Para o butano: ou (isobutano)
↳ Nomenclatura comum

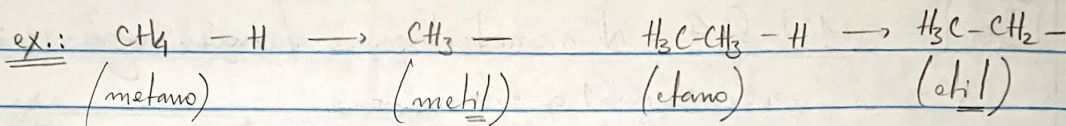
Isômeros constitucionais!

• Quanto mais carbonos, mais isômeros constitucionais
↳ Nomenclatura sistemática (IUPAC)
Cadeia parental + substituintes (?)

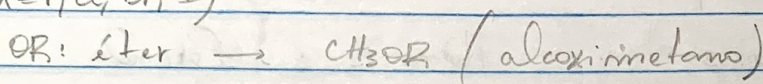
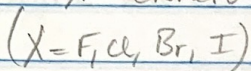
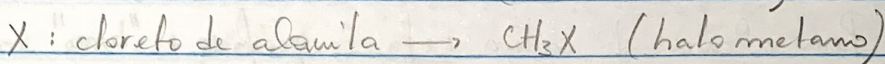
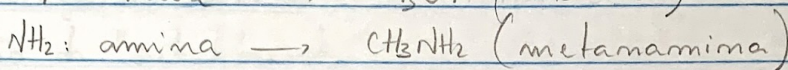
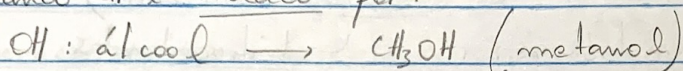
⇒ Nomenclatura de substituintes

Alcano - H → Substituinte alquil (R)
(prefixo + "ano") (prefixo + "il")

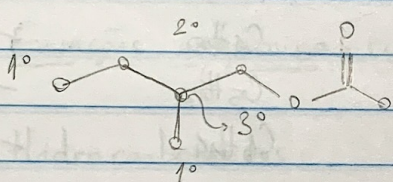
2



Quando H é trocado por:



Obs.: Tipos de carbono:

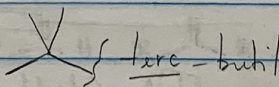
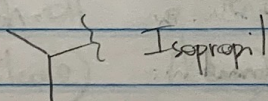
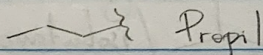
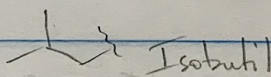
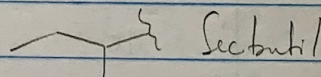
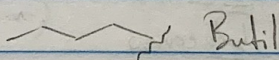
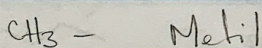


1° : ligado a 1 carbono

2° : " " 2 carbonos

3° : " " 3 carbonos

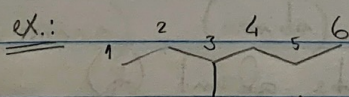
Substituintes comuns:



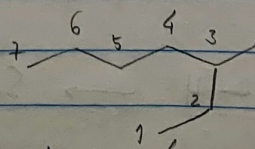
Muitas nomenclaturas comuns!

$\Rightarrow$ Regras de nomenclatura:

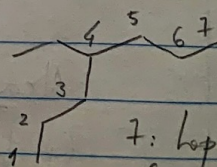
1. Determinar a cadeia carbônica mais longa: cadeia parental (nome base; último nome)



6: hexano (nome base)



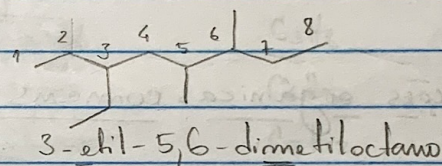
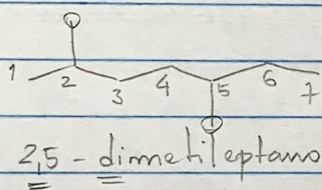
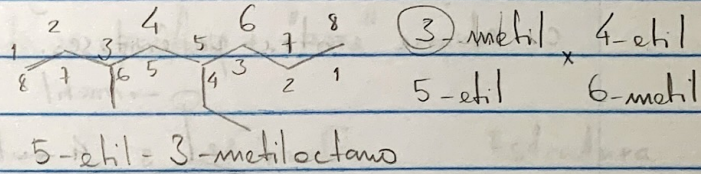
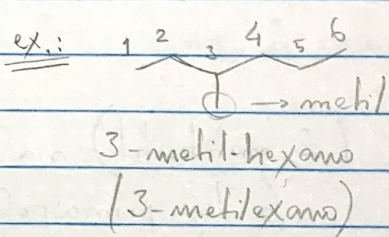
7: heptano (nome base)



8: octano (nome base)

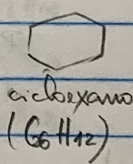
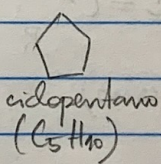
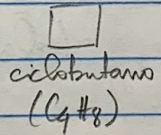
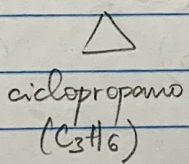
b. Número-substituinte no início do nome

- * Enumeração: menor valor possível
- * Mais de um subs.: menor dos números possíveis
- * Mais de um subs. igual: "di", "tri", "tetra" e ordem alfabética no nome

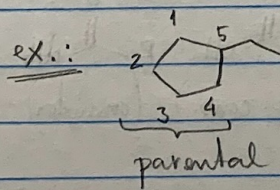


di, tri e tetra não contam para ordem alfabética

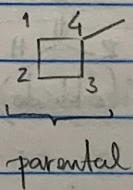
⇒ Para cicloalcanos (alcanos cíclicos): "cicl" + prefixo + "ano"



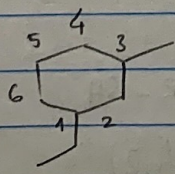
→ C_nH_{2n}
(IDH = 1!)



Etilciclo-
pentano

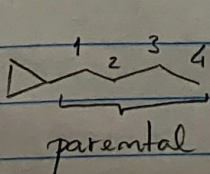


Metil ciclo-
butano



→ 1-etil-3-metilciclohexano ✓
3-etil-1-metilciclohexano x

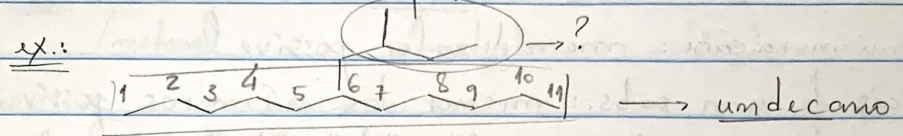
2 ou mais subs.:
enumeração



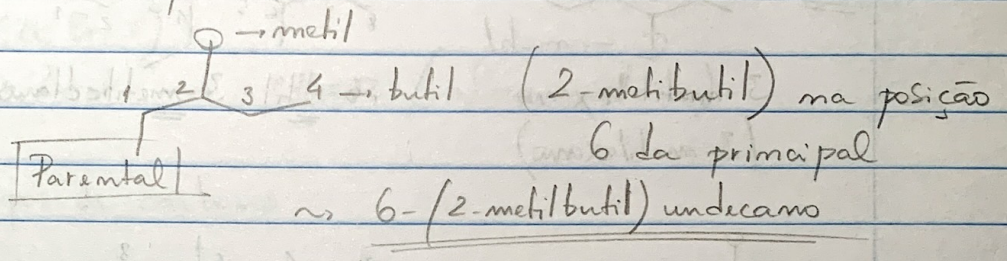
1-ciclopropil
butano

4

⇒ Substituintes complexos:

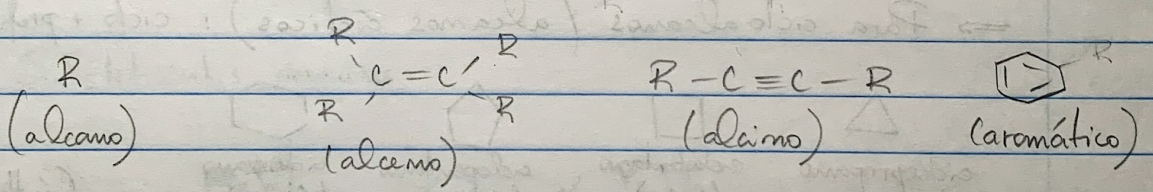


- a. Numeração a partir da cadeia parental;
- b. Posição do "substituinte";
- c. Nome entre parênteses.

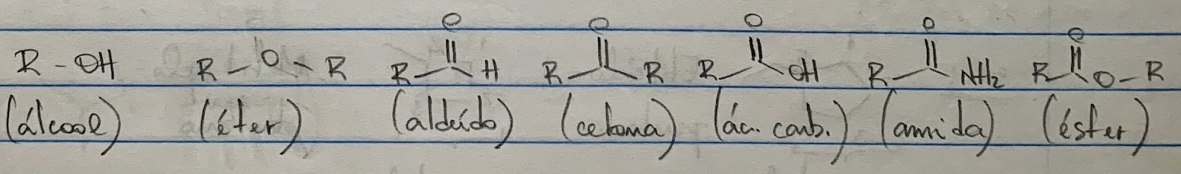


⇒ Funções orgânicas comuns:

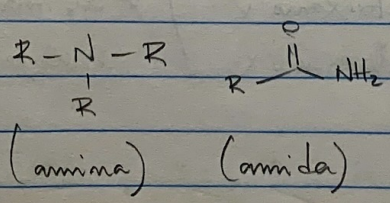
a. Hidrocarbonetos:



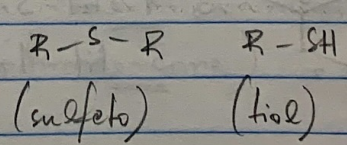
b. Oxigenados:



c. Nitrogenados:



d. Sulfurados:



e. Haletos de alquila

